PhysCapBench

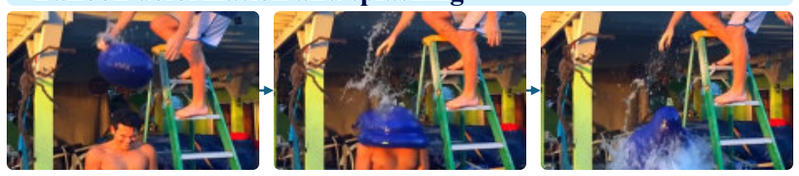
Cross-domain physical annotation, evaluation, and downstream utility

(a) Three complementary video domains

3 domains · 1,500 sequences · 500 per domain

Third-person physical scenes

Balloon deformation and splashing

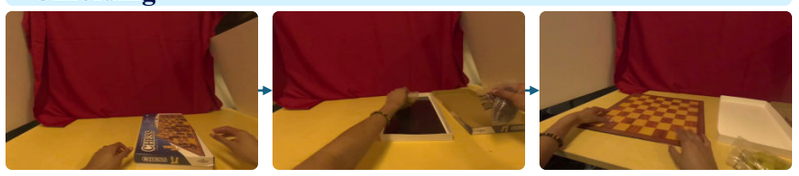


Pouring and surface pattern formation

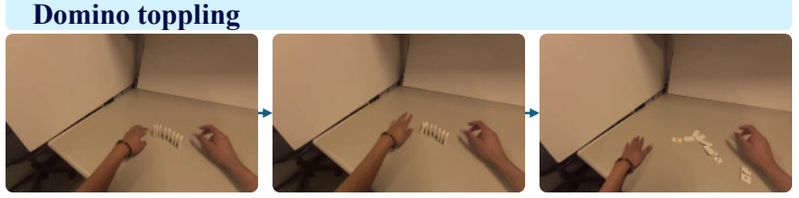


Egocentric human manipulation

Unfolding

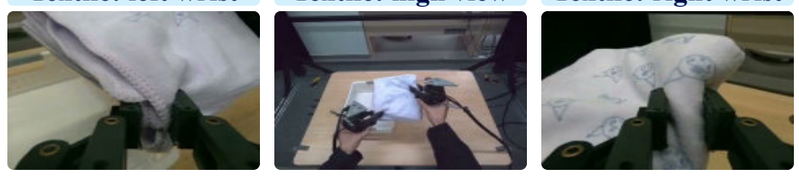


Domino toppling

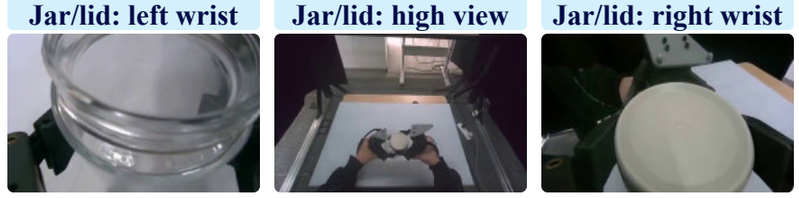


Human-operated handheld grippers

Textile: left wrist | Textile: high view | Textile: right wrist



Jar/lid: left wrist | Jar/lid: high view | Jar/lid: right wrist



Domain-adapted evidence → Shared physical representation

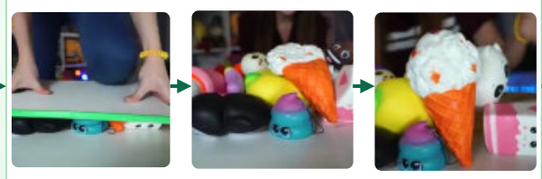
(b) Evidence-grounded physical annotation

1 Observe



RGB + domain-specific cues

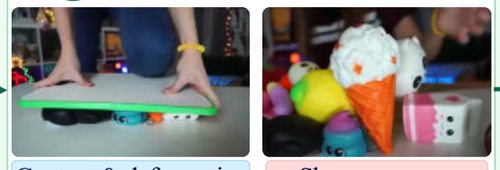
2 Structure events



Press | Release | Recover

Entities · Events · States · Relations

3 Verify evidence



Contact & deformation

Pressing deforms the soft objects.

Shape recovery

After unloading, shapes recover.

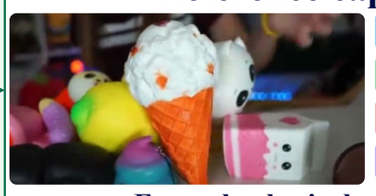
4 Human review

Correct & cross-check

Revise (if needed)

Shared description principles

Physical records & Reference captions



Entities | Events | States | Relations

Example physical caption

The board compresses the soft objects. After unloading, they recover their shapes.

(Illustrative paraphrase)

Reference Text

Physical caption

(c) Physical caption evaluation

Candidate caption + Reference Text → Text-only judge → Semantic agreement & coverage

Seven evaluation dimensions

- Event Coverage
- Objects and Roles
- Actions and Motion
- Contact and Interaction
- States and Outcomes
- Physical Mechanisms and Causality
- Temporal Order

Example candidate caption

"The board presses the toys."

Contact described

The board presses on the toys.

Outcome omitted

The toys regain their shape after release.

Reference text: caption + dimension-specific annotation text

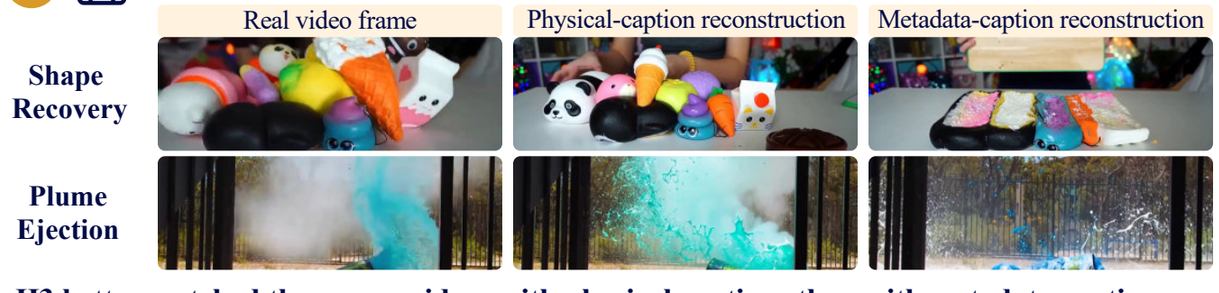
Reported evaluation: third-person + egocentric

(d) Downstream utility

1 Video reconstruction

Same first frame · Fixed generator

Same time, different caption conditions



H3 better matched the source videos with physical captions than with metadata captions.

Mean visual distance (LPIPS) fell by **9.5%** on third-person videos and **30.9%** on egocentric videos.

2 Policy instruction enrichment

RoboTwin demonstrations
Original vs. enriched instructions → Policy fine-tuning

Independent RoboTwin study

Success on one RoboTwin task

Same 40 test scenes and original test instruction

Original training text **15/40** → Enriched training text **18/40**